

Machine learning outcomes

Executive summary report for the New York City Taxi and Limousine Commission

Overview

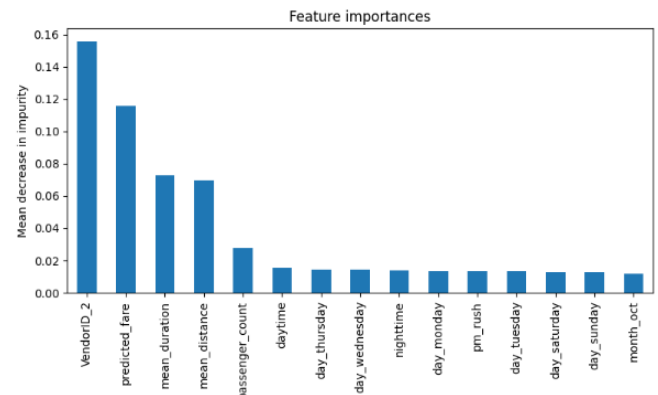
- The Automatidata data team was contracted to build a machine learning model to identify opportunities to increase taxi drivers' revenue by predicting whether a customer is likely to leave a generous tip.

Objective

- The objective is to use tree-based modeling techniques to predict the most generous customers, defined as those who are likely to tip more than 20%.
- This approach avoids the ethical concerns associated with directly limiting access to taxi services based on predicted tipping behavior, which could introduce significant risks of unfair discrimination.

Results

	model	precision	recall	F1	accuracy
0	RF CV	0.674919	0.757312	0.713601	0.680233
0	RF test	0.675297	0.779091	0.723490	0.686538



- Vendor ID, predicted fare, mean duration and mean distance were the most important features in this model.
- This model performs acceptably. Its F1 score was 0.7235 and it had an overall accuracy of 0.6865. It correctly identified ~78% of the actual responders in the test set, which is 48% better than a random guess.

Next Steps

- Test the model with a select group of taxi drivers to further evaluate its performance.
- Collect more granular driver- and user-level data, including past tipping behavior for each customer and tipping data for customers who paid in cash.
- Apply K-means clustering and analyze the resulting clusters to derive insights from the data and further refine the machine learning approach.